

ANH HOANG PHUC NGUYEN

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PROFILE SUMMARY

I am an AI/ML Engineer and Data Scientist with a Master of Artificial Intelligence and 3 years of experience. I specialise in applying AI solutions to industry products, with hands-on technical execution, particularly in the fields of computer vision and natural language processing.

SKILL

- **Programming Languages:** Python, SQL
- **AI & Data Science:** PyTorch, Tensorflow, LLMs, LoRA, RAG, LangChain, n8n
- **Cloud & Deployment:** AWS (EC2, ECR, S3, Lambda), GCP, Docker, CI/CD, Git
- **Databases:** MySQL, SQLite3, pgvector
- **Others:** Linux, Jira, Tableau, PyQt5, Flask, Apache Kafka, MQTT, RabbitMQ

WORK EXPERIENCE

AI Engineer (Sydney, Australia) May 2026 - August 2026
Chipforge.ai

- Implemented the Reflexion Loop in Chipforge platform, fix the bugs of HDL in 3-5 iterations
- Data mining and analysis for training LLM models, scraping nearly 1600 HDL Github repositories, processing the CPT, generating the HDL specifications for SFT data.

Research Assistant (Sydney, Australia) April 2025 – February 2026

University of Technology Sydney

Project: Scoliosis Assessment Application

- Built data pipeline to process and annotate 20,000 X-ray images for fine-tuning Mask-RCNN.
- Fine-tuned Mask R-CNN models for vertebrae segmentation and proposed spine curve angle measurement method, with an absolute error of $3.02^\circ \pm 2.85^\circ$.
- Deployed the system via CI/CD pipeline via GitHub Actions, Docker on AWS EC2, integrating AWS S3 for secure versioning and efficient retrieval of deep learning model weights.

Research Assistant (Melbourne, Australia) October 2025 – December 2025

The University of Melbourne

- Implemented an automated workflow in n8n that runs the data pipeline to extract and transform data from GeoScape APIs and Google Maps APIs to generate comprehensive property reports.

Machine Learning Engineer (Ho Chi Minh City, Vietnam) April 2023 - January 2024
Ycomm Viet Nam

Project: Webtoon editor – Agile development

- Deployed text inpainting API using LaMa model and text detection APIs using EasyOCR and Google Vision API, which is the main feature in the webtoon editor.
- Built a webtoon dataset pipeline for English, Korean, Japanese, and Chinese using Label Studio.

Project: Subtitle extractor – Agile development

- Deployed Stable_Whisper and Google Speech-to-text API to extract subtitles in cartoons.
- Integrated DeepL API for translation API.

Artificial Intelligence Specialist (Ha Noi and Ho Chi Minh City) October 2021 - July 2022

VinBigData

Project: Text-to-Speech with subword embedding

- Researched phoneme + subword context modelling based on Tacotron2.
- Developed a custom Vietnamese tokenizer using Hugging Face tokenization tools.
- Achieved Mel Cepstral Distortion (MCD) of 11.37, outperforming phoneme-only (12.1) and subword-only (11.77) models.

Project: Vietnamese Spelling Error Correction System

- Implemented augmentation techniques to create 5 million training samples.
- Experimented with BI-LSTM (ROUGE-1 0.86), and Transformer (ROUGE-1 0.93).

Project: Visual Question Answering

- Implemented pipeline including Faster R-CNN for image feature extraction, GloVe embeddings for question representation, and BERT for context modelling.
- Achieved overall accuracy of 70.98%.

Data Scientist (Ho Chi Minh City, Vietnam)
FPT Telecom

June 2021 - October 2021

Project: Unknown Recognition by Camera Project

- Integrated EfficientNet into the backbone of RetinaFace, reducing model weight to 50%.
- Optimized user registration using FAISS, users only provide 3 images to create account.

Artificial Intelligence Intern (Ho Chi Minh City, Vietnam)

July 2020 - October 2020

Cloud Nine Solutions

Project: Facial Recognition Check-in Project

- Developed a neural network model for real-time facial recognition.
- Deployed the solution on Jetson Nano for on-device, low-latency check-in.

EDUCATION

Master of Artificial Intelligence - University of Technology Sydney (UTS)

February 2024 - December 2025 | GPA: 6.81/7 | WAM: 93.69/100

Postgraduate Academic Excellence International Scholarship

- Participated in Waymo 3D Semantic Segmentation Challenge: trained and validated SpUNet and Point Transformer V3 on Waymo v2.0.1.
- Implemented ClearML for the Job and Candidate recommendation project, nominated for presentation at AI Showcase in UTS FEIT TechFest 2025.

Bachelor of Engineering in Computer Science - Ho Chi Minh City University of Technology (HCMUT)

August 2017 - October 2021 | GPA: 8.52 / 10.00

Certificate of Honour - the Faculty of Computer Science and Engineering.

Thesis: Biometric-Based Attendance Tracking System with One/Fewshot Learning

- Implemented Kalman filter for optimised face tracking and Apache Kafka to connect Jetson Nano and Google Cloud Platform.
- Fine-tuned FaceNet on HCMUT student dataset, achieving F1 score of 99.395% (0.335% improvement over baseline).

PERSONAL PROJECTS

Personal AI Portfolio (anhhoangphucnguyen.com)

- Built an interactive Retrieval-Augmented Generation (RAG) application using LangChain, the Claude API, and VueJS to answer recruiter queries about my background.
- Engineered a vector database backend using Supabase (PostgreSQL with pgvector) to embed and retrieve context from personal documents.
- Architected CI/CID pipeline using Docker and GitHub Actions for delivery to Google Cloud Run.

SonicJob application - an employment platform to support both job seekers and recruiters

- Collected, cleaned and combined multiple job skill datasets to fine-tune RoBERTa for Name Entity Recognition task, achieving F1 score of 81%.
- Built ClearML pipeline for data extraction, data preparation, model training, model evaluation, optimisation, and model selection.

PEER-REVIEWED PUBLICATIONS

- Evolving optimized transformer-hybrid systems for robust BCI signal processing using genetic algorithms <https://doi.org/10.1016/J.BSPC.2025.107883>

- EEG-TCNTransformer: A Temporal Convolutional Transformer for Motor Imagery Brain–Computer Interfaces. <https://doi.org/10.3390/signals5030034> ([Signals Best Paper Awards 2024](#))

CERTIFICATIONS

- Building with the Claude API
- Claude Code in Action
- Microsoft Certified: Azure AI Fundamentals